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CHEMICAL KINETICS

Reaction

Slow	fast (instantoneous)
There are reactions which occur in extremly slow rates or sometimes never happen at the room temperature	- time scale is very small
ex - graphite $\rightarrow$ diamond	ex. Human vision combusting Mg
	$Mg_{(s)} + O_{2(g)} \rightarrow MgO_{(s)}$

Moderate reactions

Between the two extremes, there is a large number of reactions which takes place at moderate and measurable rates at room temperature.

$$2H_2O_{2(aq)} \longrightarrow 2H_2O_{(l)} + O_{2(g)}$$

$$2H_{2(g)} + O_{2(g)} \longrightarrow 2H_2O_{(l)} \quad (\Delta G = -273\ kJ\,mol^{-1})$$

$$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)} \quad (\Delta H = -92\ kJ\,mol^{-1})$$

(1) Can the reaction takes place under the given condition of T and P?

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(2) To what extent will the reaction occur?

When N_2 and H_2 reaction is di equilibrium The process is occuriring. This reaction never go to completion. The yield of NH_3 is very poor.

$HCl + H_2O \longrightarrow H_3O^+ + Cl^-$ (This is a strong acid base)

$CH_3COOH + H_2O \rightleftharpoons CH_3COO^- + H_3O^+$ (This is a weak acid)

(This is at equilibrium so this reaction is not completion. Desociation is very law)

(3) How fast would the Products be formed?

Reaction is concerning viable in industrial scale. The reaction is intermediate (Not fast, not too slow)

(4) What path would the reaction take?

Thermodynamic is the answer the energy associated with the reaction. ($\Delta G, \Delta H, \Delta S$)

$\Delta G < 0$ (spontaneously)

$\Delta G = 0$ (system in equilibrium)

$\Delta G > 0$ (The reaction takes place at constant temperature & pressure)

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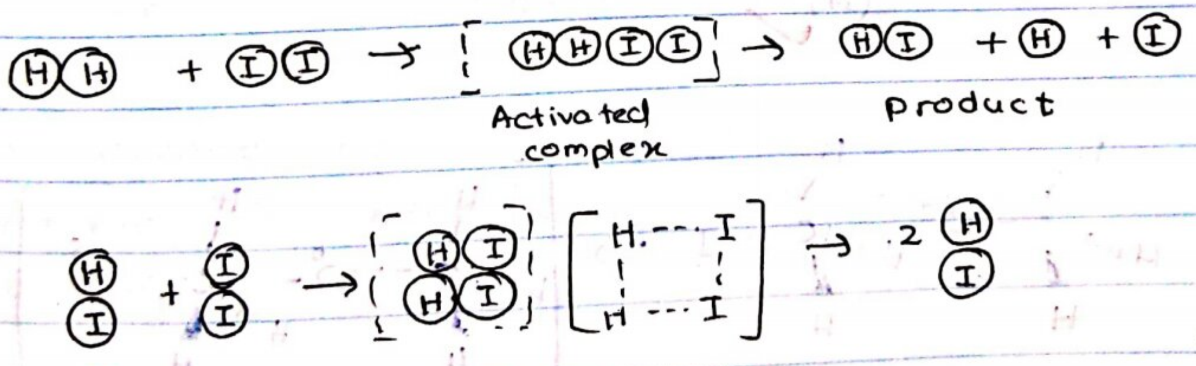
* Thermodynomic gives you only about the feasibility of the reaction. Not how much is product is formed. It tells you the reaction possible or not. How do you chonge the condition the reaction to take place.

* And the reaction kinetics tells you the rate of the reaction. How fast reaction takes place. How fast product are formed. How fast reacted are disappered.

Factors Affecting Reaction Rates

(I) molecule collide each other in favourable orientation to occur any reaction

$$H_{2(g)} + I_{2(g)} \longrightarrow 2HI_{(g)}$$



This is the favourable orientation

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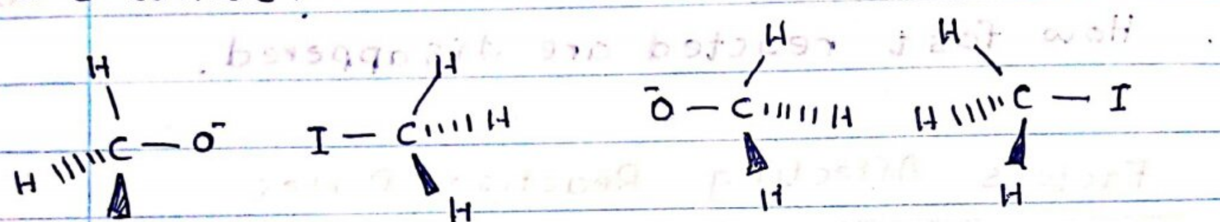
(II) collision should provide the activation energy of the reaction

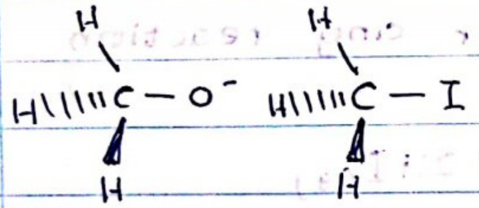
Let's consider following reaction,

$$CH_3O^- + CH_3I \longrightarrow CH_3-O-CH_3 + I^-$$

what are the possible orientations when they are collide?

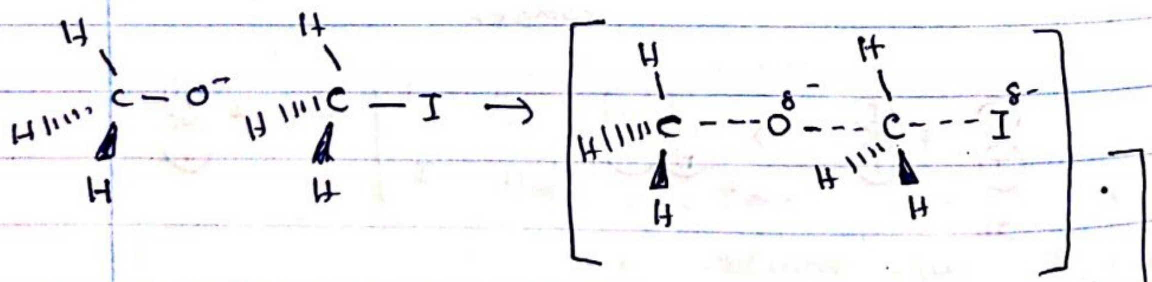


(I) (II)

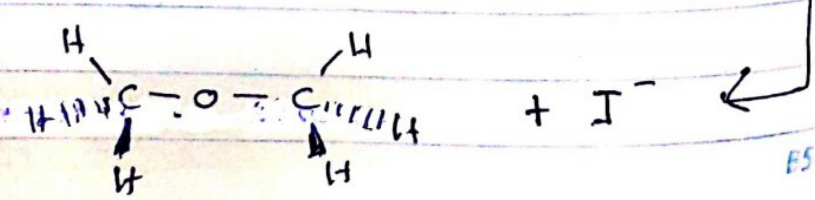


(This orientation only give product)

(III)



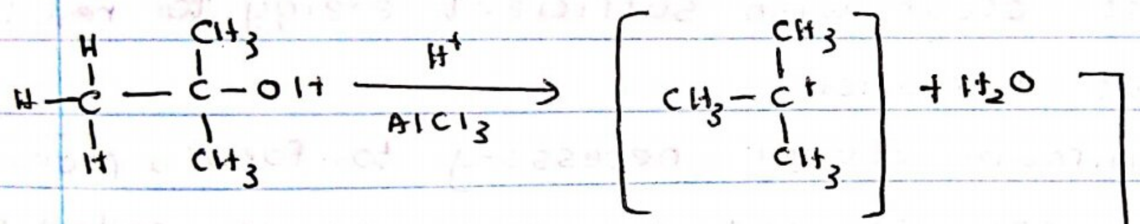
Trasition state



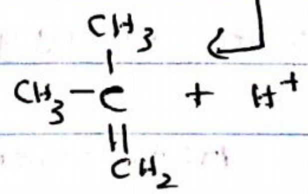
- In the transition state, the C–O and C–I are partially broken. The O–C partially formed.

This is called activated complex. This cannot be isolated. ~~There~~ In here this transition state highly unstable. Since there have partially bonds in forming & breaking. The lifetime is very short. We cannot take it out

Let's consider following reaction



In here intermediate is carbocation. If the intermediate is very stable then reaction happen very easily



You can stopped the reaction and isolate the intermediate. The lifetime is very high. There are no partially bonds in intermediate.

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Collision theory

(1) The rate of a reaction is proportional to the rate of reactant collisions:

$$\text{reaction rate} \propto \frac{\#\ \text{collisions}}{\text{time}}$$

(2) The reacting species must collide in an oreien-tation that allows contact between the atoms that will become bonded together in the product.

- In addition to a proper orientation, the collision must also occur with sufficient energy to result in product formation.
- The minimum energy necessary to form a product during a collision between reactants is called the activation energy (E_a).

Activation Energy > average kinetic Energy of the molecules ⟹ reaction occur slowly

Activation energy < averge kinetic Energy of the molecules ⟹ reaction occur rapidly

- Activated complex (transition state)

unstable combination of reactant species formed during a chemical reaction

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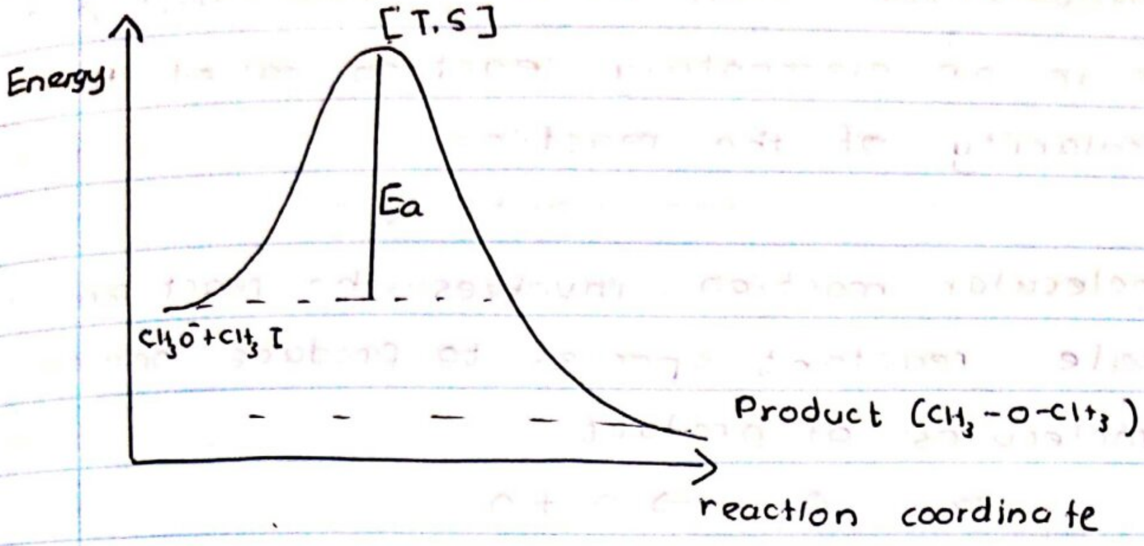
Energy Profile



- The reaction mechanism (or reaction path) provides details regarding the precise, step-by-step process by which a reaction occurs.

The decomposition of ozone,

$$2O_3 \longrightarrow 3O_2$$

$$O_3 + O_3 \longrightarrow O_3^* + O_3$$ excited molecule

$$O_3^* \longrightarrow O_2 + O$$

$$2O \longrightarrow O_2$$

- Each of the steps in a reaction mechanism is an elementary reaction.
- Elementory reaction - reaction takes place in a single step
- Species that are produced in one step and consumed in a subsequent step are called intermediates.

Molecularity - the no of atoms that partici-pate in an elementary reaction called the molecularity of the reaction.

Unimolecular reaction - involves the reaction of a single reactant species to produce one or more molecules of product.

ex:- $O_3 \longrightarrow O_2 + O$

Bimolecular elementary reactions.

- This involves two reactant species.

$$A + B \rightarrow \text{products}$$

and

$$2A \rightarrow \text{products}$$

ex:-

$$NO_{2(g)} + CO_{(g)} \longrightarrow NO_{(g)} + CO_{2(g)}$$

Termolecular Elementary reactions

An elementary termolecular reaction involves the simultaneous collision of 3 atoms, molecules or ions. This reactions are uncommon because the probobility of 3 particles colliding simultan-aneously is less than one-thousandth of the probability of two particles colliding

ex:-

$$2NO + O_2 \longrightarrow 2NO_2$$

$$\text{rate} = k[NO]^2[O_2]$$

$$2NO + Cl_2 \longrightarrow 2NOCl$$

Q. 47. Bring out the points of difference between order and molecularity of a reaction.

(Lucknow, 2000; Nagpur 2002; Purvanchal 2002; Calcutta, 2007)

Ans. The main points of difference between the order and the molecularity of a reaction are as under :

Order of reaction	*Molecularity of reaction*
1. It is the sum of the concentration terms on which the rate of reaction actually depends or it is the sum of the exponents of the concentrations in the rate law equation.	1. In simple reactions, it is equal to the number of molecules of the reactants as represented in the balanced equation. In complex reactions, it is the number of molecules involved in the slowest step.
2. It need not be a whole number *i.e.* it can be fractional as well as zero.	2. It is always a whole number.
3. Order of even simple reaction *may not* be equal to the number of molecules of the reactants as seen from the balanced equation.	3. For simple reactions,, the molecularity can usually be obtained from the stoichiometry of the equation.
4. It can be determined experimentally only and cannot be calculated.	4. It can be calculated by simply adding the molecules of the slowest step.

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- Elimentary reactions are very limit. The most of the reactions are occuring as multisteps reactions.

$$H_{2(g)} + O_{2(g)} \longrightarrow H_2O$$

This reaction is not happen in one step. And the activation energy is very high. The energy produced in collision atoms is not enough to reached the activation energy to form H_2O. Thus, we have to give energy to access the activation energy.

$$\left.\begin{array}{l} H_{2(g)} + I_{2(g)} \longrightarrow 2HI_{(g)} \\ I_{2(g)} \rightleftharpoons I_{(g)} + I_{(g)} \\ H_{2(g)} + I_{(g)} + I_{(g)} \longrightarrow 2HI_{(g)} \end{array}\right\} \text{This is a multi step reaction}$$

One step in a multistep reaction mechanism is significanty slower that the others. This step will limit the rate at which the overall reaction occurs.

The slowest step is therefore called the <u>rate-limiting step</u>.

The factors affect the rate

(i) Concentration of reactants

(ii) Temperature

(iii) catalyst

(v) Electromagnetic radiation (UV, IR)

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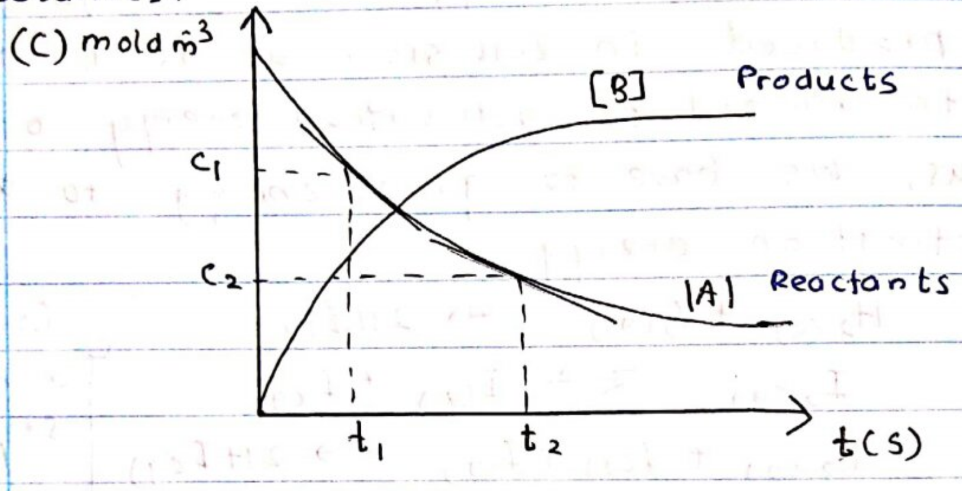
(1) Effect of concentrations

- With the exception of zero order reactions, the rate of every chemical reaction is directly proportional to the concentrations of the reacting substances.



- Higher the molecules in volume (concentration) is high. Therefore no of collisions in that volume is high. Then rate is increased.

$$A \longrightarrow B$$

Rate of consumed [A] $= \dfrac{[A_2]-[A_1]}{t_2-t_1} = \dfrac{-\Delta[A]}{\Delta t}$

Rate of formed of [B] $= \dfrac{\Delta[B]}{\Delta t}$

(The negative sign appears since the concentration is decreasing with time)

Instantaneous rate - the rate of the instant of interest

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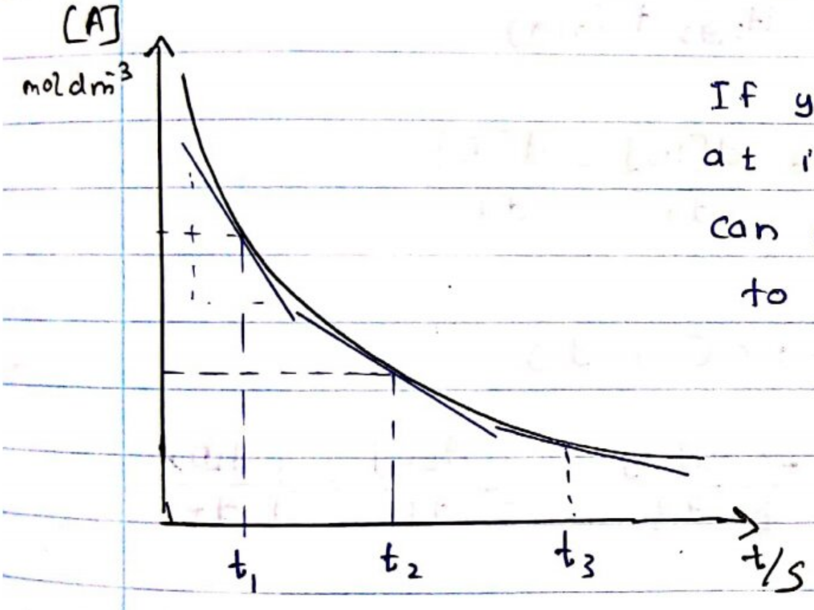
The rate of formation of a product with respect to time

Rate of change in concentration of a reactant with respect to time denoted by,

$$\frac{dc}{dt} = \lim_{\delta t \to 0} \frac{\delta c}{\delta t}$$

- how fast reactant disappear

$$-\frac{dC}{dt}$$



If you want to get rate at instant time you can draw a tangent line to the point and get slope

$$A + B \longrightarrow C + D$$

At a given temperature,

Rate $\propto$ the produt of concentration of each reactant.

Rate $\propto [A]^{\alpha} [B]^{\beta}$

Rate $= k [A]^{\alpha} [B]^{\beta}$

rate constant

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* The order of a reaction is defined as the sum of the powers of the concentration terms in the equation.

α = order of the reaction with respect to A

β = " " " " " B

$(\alpha + \beta)$ = overall order of the reaction

$$2HI_{(g)} \longrightarrow H_{2(g)} + I_{2(g)}$$

$$\text{Rate} = -\frac{1}{2}\frac{d[HI]}{dt} = \frac{d[H_2]}{dt} = \frac{d[I_2]}{dt}$$

$$aA + bB \longrightarrow cC + dD$$

$$\text{Rate} = -\frac{1}{a}\frac{d[A]}{dt} = -\frac{1}{b}\frac{d[B]}{dt} = \frac{1}{c}\frac{d[C]}{dt} = \frac{1}{d}\frac{d[D]}{dt}$$

* The order has to be experimentally determined.

* It is not necessary that the order of a reaction should be a whole number but can be zero or fractional. There are certain reactions, mostly heterogeneous reactions, for which the rate is zero or fractional.

* For elimentry reaction the order is of each reactants is equall to the stoichiometric constant of balance equation.

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* Other chemical equations we don't know whether it is elementary or complex reaction. So we can't determine the order without experiment.

Zero order reactions

- In zero order reactions the concentration of the reactants remain constant during the course of the reaction.
- The rate is independent of the concentration of such reacting molecules.

$$A \longrightarrow P$$

$$\text{Rate} = \frac{-d[A]}{dt} = k[A]^0$$

$$= \frac{-d[A]}{dt} = k$$

$$-d[A] = k\,dt$$

$$-\int_{C_0}^{C_t} d[A] = k\int_{t_0}^{t} dt$$

C_0 - initial concentration of A at 't_0'

C_t - final " " " A at 't'

$$-[A]_{C_0}^{C_t} = k[t]_{t_0}^{t}$$

$$C_t - C_0 = -kt$$

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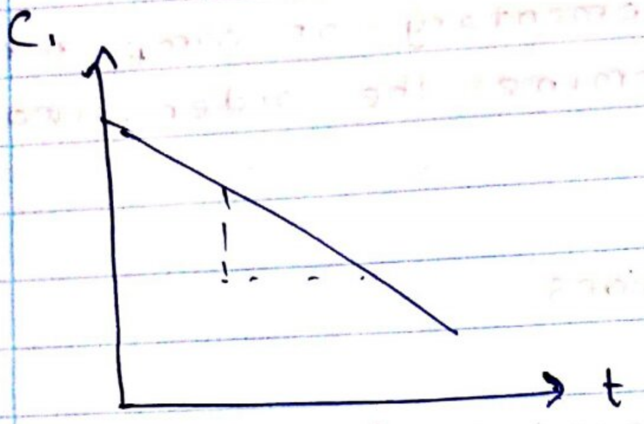
$$C_t = -kt + C_0$$

$$y = -mx + C$$

(units of $k = mol\,dm^{-3}\,s^{-1}$)



* For a zero order reaction, concentration of product increases with increase of reaction time

Half Life of the zero order reaction ($t_{1/2}$)

- Time taken for the initial concentration to be half

$$C_t = C_0/2$$

$$-\int_{C_0}^{C_0/2} d[A] = k\int_{t_0}^{t_{1/2}} dt$$

$$-[A]_{C_0}^{C_0/2} = k[t]_0^{t_{1/2}}$$

$$-\left[C_0/2 - C_0\right] = k\left[t_{1/2} - 0\right]$$

$$\frac{C_0}{2} = -k\,t_{1/2}$$

$$\boxed{t_{1/2} = \frac{C_0}{2k}}$$

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Examples of zero order reaction

~~Photochemical~~ (1) reaction between $H_{2(g)}$ and $Cl_{2(g)}$ in the presence of sunlight

(2) the decomposition of hydrogen iodide and NH_3 at the surface of gold (Au)

First ORDER REACTIONS

$A \longrightarrow P$ $\boxed{\text{units of } k = s^{-1}}$

$$\text{Rate} = \frac{-d[A]}{dt} = k[A]$$

$$\frac{-d[A]}{[A]} = k\,dt$$

$$\int_{C_0}^{C_t} \frac{d[A]}{[A]} = -k\int_{t=0}^{t} dt$$

$$\ln[A]_{C_0}^{C_t} = -k[t]_0^t$$

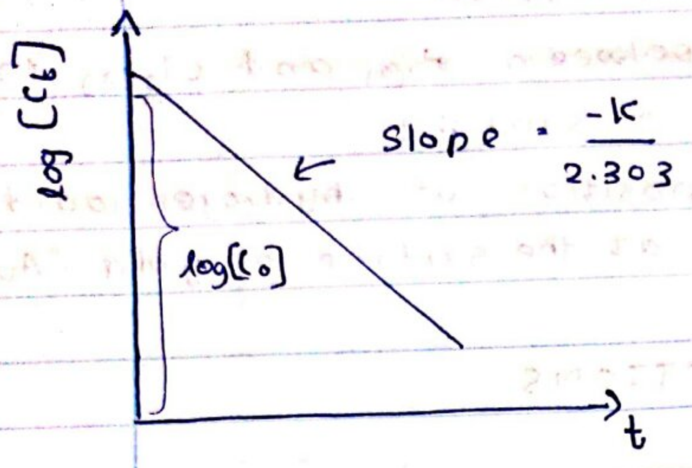
$$\ln[C_t] - \ln[C_0] = -kt$$

$$\ln[C_t] = -kt + \ln[C_0]$$

$$\cancel{\ln[C_t]} = \cancel{-kt}$$

$$2.303\log[C_t] = -kt + 2.303\log[C_0]$$

$$\log[C_t] = \frac{-kt}{2.303} + \log[C_0]$$



Half life of the first order reaction

$$\int_{C_0}^{C_0/2} \frac{d[A]}{[A]} = -k\int_0^{t_{1/2}} dt$$

$$\ln[A]_{C_0}^{C_0/2} = -k[t]_0^{t_{1/2}}$$

$$\ln[C/2] - \ln[C_0] = -k[t_{1/2}]$$

$$t_{1/2} = \frac{1}{k}\ln\left[\frac{C_0/2}{C_0}\right]$$

$$\boxed{t_{1/2} = \frac{0.693}{k}}$$

Examples :-

(a) The decomposition of nitrogen pentoxide

$$2N_2O_{5(g)} = 4NO_{2(g)} + O_{2(g)}$$

(b) Hydrolysis of methyl acetate in aqueous solution

$$CH_3COOCH_3 + H_2O \xrightarrow{H^+} CH_3COOH + CH_3OH$$

$$-\frac{d[CH_3COOCH_3]}{dt} = k\, C_{CH_3COOCH_3}\, C_{H_2O}$$

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Second order Reactions

$$A \longrightarrow P$$

$$-\frac{d[A]}{dt} = k[A]^2 \quad \leftarrow \text{differential rate}$$

$$\frac{d[A]}{[A]^2} = -k\,dt$$

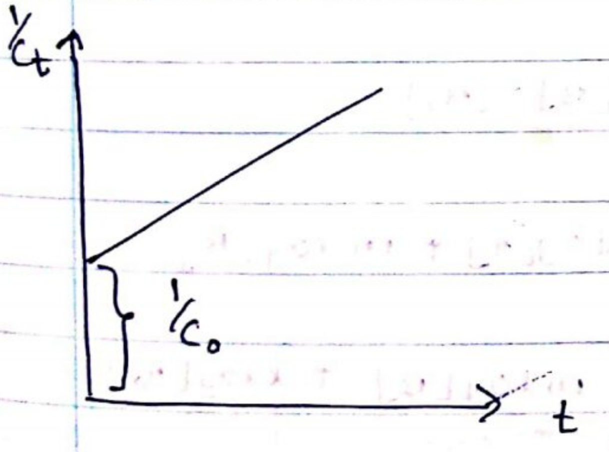
$$\int_{C_o}^{C_t} \frac{d[A]}{[A]^2} = -k \int_0^t dt$$

$$-\left[\frac{1}{[A]}\right]_{C_o}^{C_t} = -k[t]_0^t$$

$$-\left(\frac{1}{C_t} - \frac{1}{C_o}\right) = -k[t-0]$$

$$\frac{1}{C_t} - \frac{1}{C_o} = kt$$

$$\frac{1}{C_t} = kt + \frac{1}{C_o} \quad \leftarrow \text{integration rate}$$



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Half life of Second order

$$\int_{C_0}^{C_0/2} \frac{d[A]}{[A]^2} = k \int_0^{t_{1/2}} dt$$

$$\left[\frac{1}{[A]}\right]_{C_0}^{C_0/2} = k\left[t_{1/2} - 0\right]$$

$$\left[\frac{2}{C_0} - \frac{1}{C_0}\right] = k\, t_{1/2}$$

$$\boxed{t_{1/2} = \frac{1}{C_0 k}}$$

Initial Rate Method :

$$A + B \longrightarrow P$$

$$\text{rate} = \frac{-d[A]}{dt} = \frac{-d[B]}{dt} = \frac{d[P]}{dt}$$

$$\frac{-d[A]}{dt} = k[A]^{l}[B]^{m}$$

at $t=0$, $[A] = [A_0]$, $[B] = [B_0]$

$$\log\left(\frac{d[A]}{dt}\right) = \log k + l \log[A] + m \log[B]$$

$$\log\left(\frac{d[A]}{dt}\right) = \underbrace{\log k + m \log[B]}_{C} + \underbrace{l}_{m} \underbrace{\log[A_0]}_{x}$$

$$y = mx + c$$

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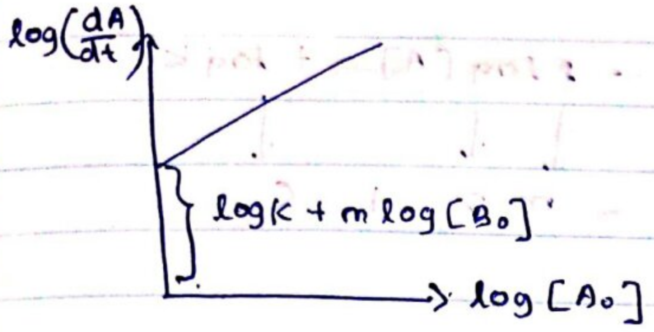

$$\underbrace{\log\left[\frac{d[A]}{dt}\right]}_{y} = \underbrace{m}_{m}\ \underbrace{\log [B_0]}_{x} + \underbrace{\log k + l \log [A_0]}_{c}$$

* By the intercept we can find 'l' and 'm'

Isolation Method

$$A + B \longrightarrow P$$

$$\frac{-d[A]}{dt} = k[A]^l[B]^m$$

In this method one of the reactant is kept in large excess, then its concentration is almost constant, and therefore, rate equation reduces to.

$$[B] >>>> [A]$$

$$\text{Rate} = \frac{-d[A]}{dt} = k'[A]^l$$

* The rate of the reaction now, is determined only by the concentration of the 'A'.
* Reactions of this type are known as Pseudo-order reactions since the order has been reduced.

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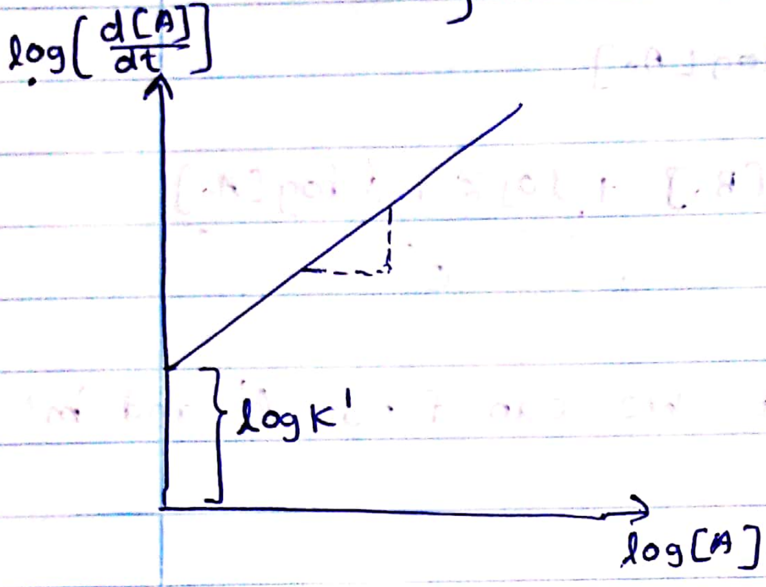
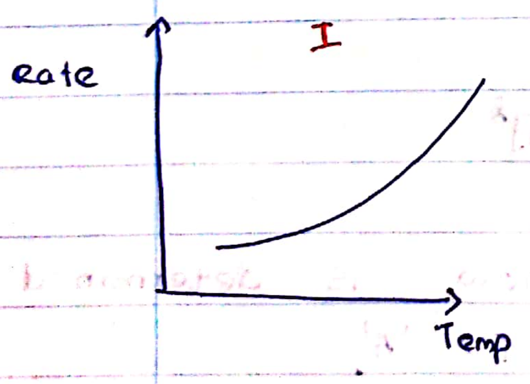
$$\log\left[\frac{d[A]}{dt}\right] = \ell \log[A] + \log k'$$

$$y = m\,x + c$$

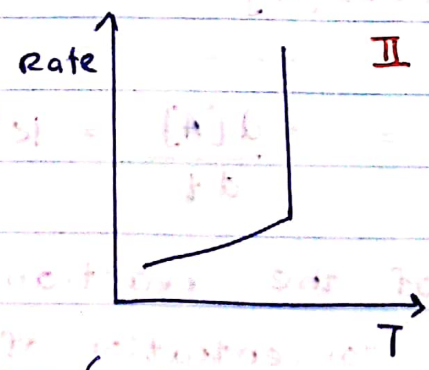


(II) Effect of TEMPERATURE

- The temperature has got a significant effect on the rate of the reaction.
- The increasing in temperature leads to a remarkable increase in the rate of a chemical reaction and hence in rate constants.

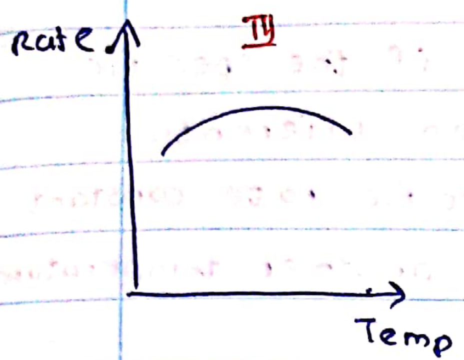


This curve is obtained most of the reaction

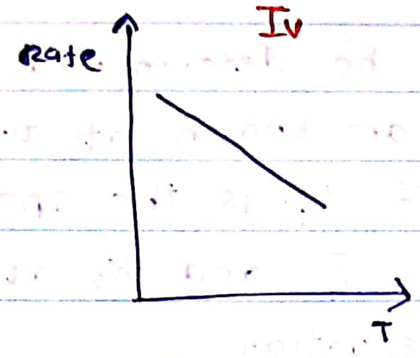


(For explosions)

In this curve, a sudden rise in the rate of the reaction is shown.



(In catalytic hydrogenations and in enzyme reactions)



(This curve is obtained only for the reaction $2NO + O_2 \rightarrow 2NO_2$)

* For homogeneous reactions, the rate is doubled for every 10°C rise of temperature. (Type I)

* Arrhenius extended his ideas and suggested a equation to show the variation of rate constant with temperature.

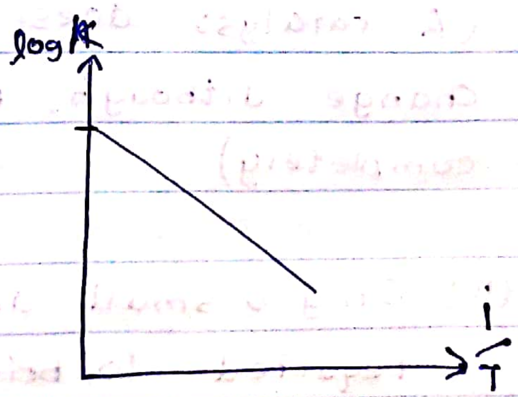
$$k = Ae^{-Ea/RT}$$

A - frequency factor

Ea - Activation energy

$$\boxed{\log k = \frac{-Ea}{2.303RT} + \log A}$$

$$y = -mx + c$$



Ans. The effect of temperature on the rate of a reaction was studied by Arrhenius. The equation, called **Arrhenius equation,** is usually written in the form

$$k = Ae^{-E_a/RT} \quad ...(i)$$

where A is a constant called **frequency factor** (because it gives the frequency of collisions of the reacting molecules), E_a is called **energy of activation,** R is gas constant and T is the absolute temperature.

The energy of activation (E_a) is an important quantity as it is characteristic of the reaction. Using the above equation, its value can be calculated as follows :

Taking logarithm of both side of equation (i), we get

$$\ln k = \ln A - \frac{E_a}{RT} \quad ...(ii)$$

If the values of the rate constant at temperatures T_1 and T_2 are k_1 and k_2 respectively, then we have

$$\ln k_1 = \ln A - \frac{E_a}{RT_1} \quad ...(iii)$$

and

$$\ln k_2 = \ln A - \frac{E_a}{RT_2} \quad ...(iv)$$

Subtracting equation (iii) from equation (iv), we get

$$\ln k_2 - \ln k_1 = -\frac{E_a}{RT_2} - \left(-\frac{E_a}{RT_1}\right)$$

$$= \frac{E_a}{RT_1} - \frac{E_a}{RT_2}$$

or

$$\ln \frac{k_2}{k_1} = \frac{E_a}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

$$= \frac{E_a}{R}\left(\frac{T_2 - T_1}{T_1 T_2}\right)$$

or

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303R}\left(\frac{T_2 - T_1}{T_1 T_2}\right) \quad ...(v)$$

Thus knowing the values of the rate constants k_i and k_2 at two different temperatures T_1 and T_2, the value of E_a can be calculated. Alternatively, knowing the rate constant at any one temperature, its value at another temperature can be calculated provided the value of E_a is known for that reaction.

Equation (ii), may be written as

$$\ln k = -\frac{E_a}{RT} + \ln A$$

$$\log k = -\frac{E_a}{2.303\,RT} + \log A \quad ...(vi)$$

This equation is of the form $y = mx + c$ *i.e.* the equation of a straight line. Thus a plot of log k *vs* $\frac{1}{T}$ is a straight line as shown in Fig. 4.8.

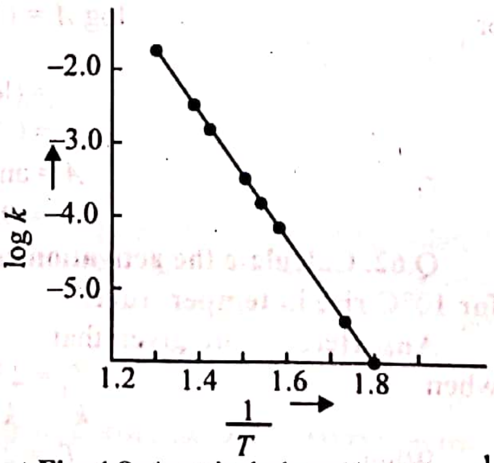


Fig. 4.8. A typical plot of log k *vs.* $\frac{1}{T}$

(iii) Effect of CATALYSIS

A catalyst is a substance which alters the speed of a chemical reaction without undergoing any chemical change and can be recoved at the end of the reaction

General Characteristics of CATALYZED REACTIONS

(i) The catalyst remains unchanged in chemical composition at the end of the reaction.
(A catalyst does not undergo any chemical change although, its physical form may change completely)

(ii) Only a small amount of the catalyst is required to bring about a reaction.

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(There are some homogeneous catalyzed reactions where the rate of the reaction increases with the increase in the concentration of the catalyst.
(ex:- The inversion of cane sugar in presence of HCl)

(iii) A catalyst does not start or initiate a reaction, but it merely increases or decreases its speed.

(iv) Catalysis does not change enthalpy of the reaction and equilibrium constant.
(A catalyst accelerates equally the rates of both forward and reversible reactions and helps to establish equilibrium quickly)

(v) A catalyst is specific in action.
(This implies that a given catalyst can catalyze only particular reaction and cannot be used to bring about every reaction)

(vi) Catalysis provides an alternative path which avoids - the slow - rate determines step of an uncatalysed reaction, It changes the mechanism of the reaction

(VII) The activity of a catalyst is enhanced by the presence of a substance called Promoter. (Ex:- in Haber's process for the manufacture of NH_3, molybdenum is used as a promoter to the catalyst iron)

(VIII) A catalyst is poisoned by the presence of small quantities of certain substances called catalytic poisons.

(ex:- arsenious oxide, HCN, CO)

the duty of the catalyst inactive.

Types of CATALYSIS

(1) Homogeneous catalysis

- the catalyst is present in the same phase as the reactants. There can be gaseous or liquid phase.

ex:-

In gas phase - very rare.

$$2SO_{2(g)} + O_{2(g)} \xrightarrow{NO} 2SO_{3(g)}$$

$$2CO + O_2 \xrightarrow{NO} 2CO_2$$

$$2N_2O \xrightarrow{NO} 2N_2 + O_2$$

(2) Heterogeneous catalysis:

- catalyst constitutes a seperote phase from the reactions. The catalyst is generally solid and reactants are mostly gases and sometimes liquids.

ex:-

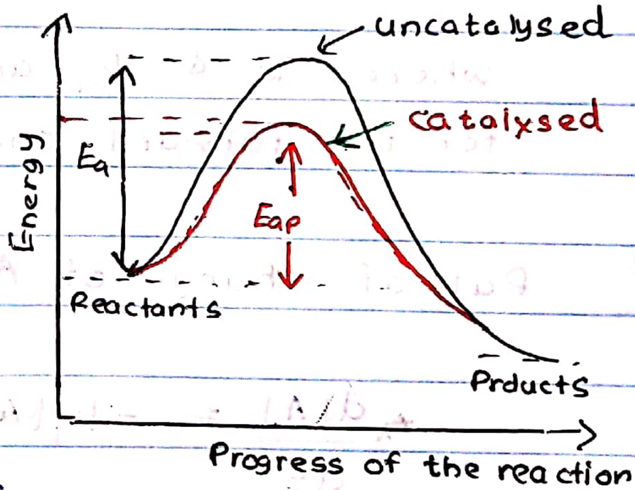
$$N_{2(g)} + 3H_{2(g)} \xrightarrow{Fe} 2NH_{3(g)}$$

$$2SO_{2(g)} + O_{2(g)} \xrightarrow{Pt} 2SO_{3(g)}$$

$$2H_{2(g)} + O_2 \xrightarrow{Pt} 2H_2O$$

NOTE :

The presence of a catalyst provides an alternate path with lower energy barrier. Thus, the energy of activation is lowered and hence a greater number of molecules



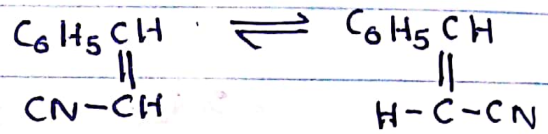
Can cross the barrier. and change over to products, thereby increasing rote of the reaction.

COMPLEX REACTIONS

(1) Reversible First order Reactions

Reactions proceeds on both directions and at equilibrium. The rate of forward reaction is equal to that of reverse reaction. Net rate of formation of each species is zero.

$$A \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} B$$



(cis - trans isomerism)

where k_1 & k_{-1} are the first order rate constants for the forward and reverse reactions respectively.

Rate of change of A

$$\frac{d[A]}{dt} = -k_1[A] + k_{-1}[B] \quad \text{—①}$$

at $t = 0$ $[A] = [A_0]$; $[B] = 0$

at $t = t$ $[A] = [A]$ $[B] = [B]$

$$[B] = [A_0] - [A] \quad \text{—②}$$

From ①

$$-\frac{d[A]}{dt} = -k_1[A] + k_{-1}([A_0] - [A])$$

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$$-\frac{d[A]}{dt} = k_{-1}[A_0] - (k_1+k_{-1})[A] \quad —③$$

At equilibrium, the two rates are equal.

$$\frac{d[A]}{dt} = 0 \qquad [A] = [A]_{equl}$$

$$k_{-1}[A_0] = (k_1 + k_{-1})[A]_{equal}$$

$$[A_0] = \left(\frac{k_1 + k_{-1}}{k_{-1}}\right)\cdot[A]_{equal} \quad —④$$

④ in ③

$$-\frac{d[A]}{dt} = k_{-1}\left(\frac{k_1+k_{-1}}{k_{-1}}\right)[A]_{equal} - (k_1+k_{-1})[A]$$

$$-\frac{d[A]}{dt} = -(k_1+k_{-1})\left([A] - [A]_{equal}\right)$$

$$-\frac{d[A]}{[A]-[A]_{equal}} = (k_1+k_{-1})\,dt$$

Integration ; at $t=0$, $[A] = [A]_0$

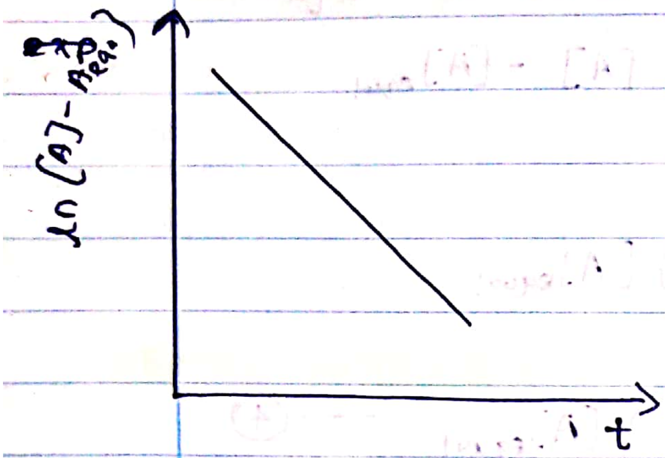
$t = t$, $[A] = [A]$

$$\cancel{\ln\frac{[A_0]-[A]_{eq}}{[A]-[A]_{eq}} = (k_1+k_{-1})\,t}$$

$$\ln\left(\frac{[A]-[A]_{eq}}{[A]_0-[A]_{eq}}\right) = -(k_1+k_{-1})\,t \quad —⑤$$

$$\ln\left[[A] - [A]_{equa}\right] = -(k_1 + k_{-1})t + \ln\left[[A]_0 - [A]_{equi}\right]$$

$$y = -\,m\,x + c$$



$$\text{Slop} = -(k_1 + k_{-1})$$

At equealibrium, two rates are equal ~~to~~ ~~zero~~ therefore

$$\frac{k_1}{k_{-1}} = \frac{[B]_{equal}}{[A]_{equal}} = \frac{[A]_0 - [A]_{eq}}{[A]_{eq}}$$

$$[A]_{eq} = \frac{k_{-1}}{k_1 + k_{-1}}[A]_0$$

$$\frac{\Delta[A]}{\Delta[A_0]} = \ln\left[-(k_1 + k_{-1})t\right]$$

$$\Delta[A] = \Delta[A_0]\, e^{-(k_1 + k_{-1})t}$$

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(2) Consecutive first-order Reactions

* There are those reactions in which the product of one reaction becomes the reactant for the other reaction.

$$A \xrightarrow{k_1} B \xrightarrow{k_2} C$$

* In general, if a reaction takes place in series, then the slowest reaction will be the rate-determining reaction.

$$\frac{d[A]}{dt} = -k_1[A] \quad \text{— (1)}$$

$$\frac{d[B]}{dt} = k_1[A] - k_2[B] \quad \text{— (2)}$$

$$\frac{d[C]}{dt} = k_2[B] \quad \text{— (3)}$$

at $t=0$, $[A] = [A_0]$, $[B] = [C] = 0$

Integration of equation (1),

$$\int_{A_0}^{A_t} \frac{d[A]}{[A]} = -k_1 \int_0^t dt$$

$$\ln[A]_0 - \ln[A]_t = k_1 t$$

$$\ln[A]_t = \ln[A_0] - k_1 t$$

$$[A]_t = [A_0]\, e^{-k_1 t} \quad \text{— (4)}$$

where, $[A_0]$ = initial concentration of A

To get concentration of B at time = t :

$$\frac{d[B]}{dt} = k_1[A_0]e^{-k_1 t} - k_2[B]$$

at $t=0$, $[B]=0$ and at time t, $[B]=[B]$

$$[B] = \frac{k_1[A_0]}{k_2-k_1}\left[e^{-k_1 t} - e^{-k_2 t}\right] \quad \text{—⑤}$$

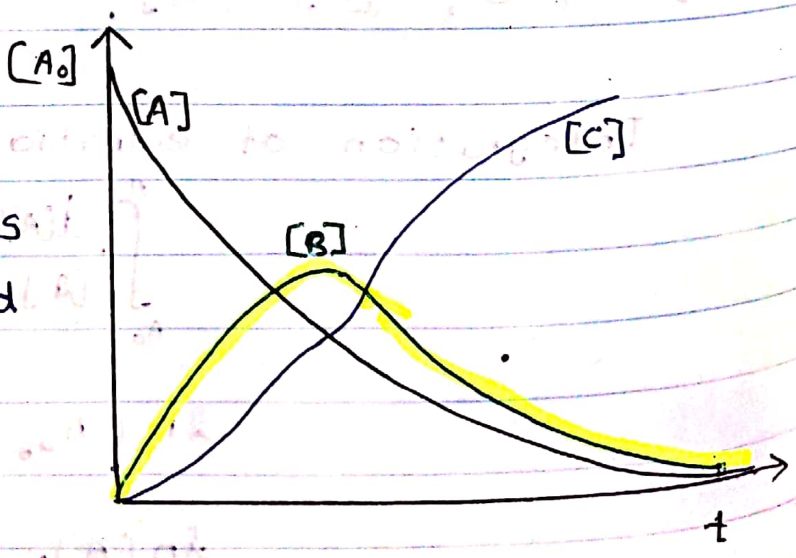
To find [C], using the conservation of matter, the total no of moles present in constant with time

$$[A_0] = [A] + [B] + [C]$$

$$[C] = [A_0] - [A] - [B]$$

$$[C] = [A_0]\left(1 - \frac{k_2}{k_2-k_1}e^{-k_1 t} + \frac{k_1}{k_2-k_1}e^{-k_2 t}\right) \quad \text{— ⑥}$$

* This graph shows that intermediate concentration [B] rises to a maximum and then drops back to zero as reactant [A] is depleted



Suppose $k_2 >>> k_1$, then whenever B is formed it decays quickly to C. The rate of formation of the product C is then determined by the rate at which [B] is formed

$$e^{-k_1 t} >> e^{-k_2 t}$$

Then we can approximate equation ⑥

$$[C] = [A_0](1 - e^{-k_1 t})$$

k_1 is the rate constant of the slowest step. B is highly reactive intemediate and equation ⑤ becomes

$$[B] = \frac{k_1}{k_2}[A_0]e^{-k_1 t}$$

$$= \frac{k_1}{k_2}[A]$$

When A is reacting only slowly, therefore, within a short-period of time. [A] can be assumed to be a constant and k_1/k_2 is very small.

The concentration of [B] would be very small at any given time when compared to [A]

So that $\frac{d[B]}{dt} = 0$

[B] is constant.

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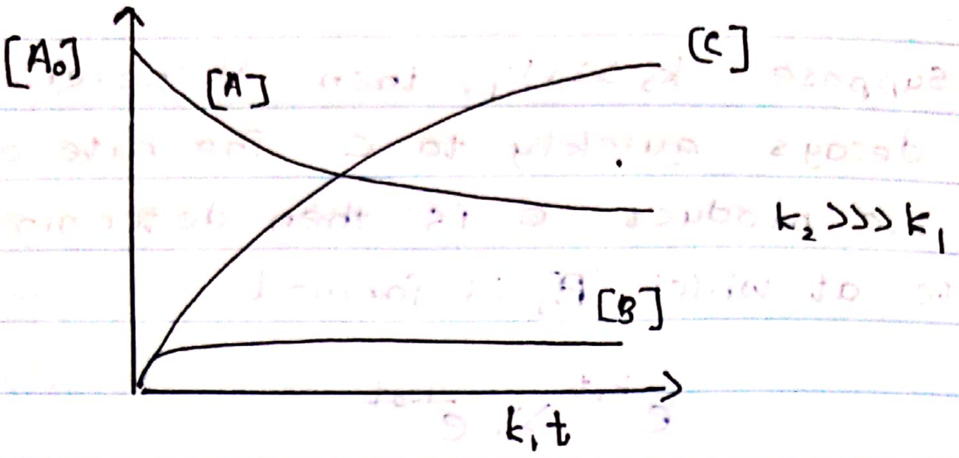

NOTE !

$\boxed{\text{In multistep reactions the concentration of intermediate can be neglect}}$

The steady - state Approximation

In multistep reactions mechanisms one or more intermediate species that do not appear in the overall equation are very reactive. and therefore do not accumulate to any significant extent during the reaction. So we can assume the concentration of [B] rises to maximum then fall back to zero.

Steady state approximation assumes that, except for initial period of time the rate of formation of the reaction intermediate essentially equals its rate of විනාශය distruction. So as to keep it at a near constant steady - state concentration.

E.g: The rate law for the Br^- catalysed aqueous reaction

$$H^+ + HNO_2 + C_6H_5NH_2 \xrightarrow{Br^-} C_6H_5N_2^+ + 2H_2O$$

The proposed mechanism is

$$H^+ + HNO_2 \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} H_2NO_2^+ \quad \text{(fast equation)}$$

$$H_2NO_2^+ + Br^- \xrightarrow{k_2} ONBr + H_2O \quad \text{(slow)}$$

$$ONBr + C_6H_5NH_2 \xrightarrow{k_3} C_6H_5N_2^+ + H_2O + Br^- \quad \text{(fast)}$$

The observed rate law is,

$$r = k[H^+][HNO_2][Br^-] \quad \text{—(1)}$$

$$\text{rate} = \frac{d[C_6H_5N_2^+]}{dt} = k_3[ONBr][C_6H_5NH_2] \quad \text{—(2)}$$

Intermediates are $H_2NO_2^+$ and $ONBr$

To eliminate $ONBr$, apply the S. state Approximation

$$\frac{d[ONBr]}{dt} = k_2[Br^-][H_2NO_2^+] - k_3[ONBr][C_6H_5NH_2] = 0$$

$$k_2[Br^-][H_2NO_2^+] = k_3[ONBr][C_6H_5NH_2]$$

$$[ONBr] = \frac{k_2[Br^-][H_2NO_2^+]}{k_3[C_6H_5NH_2]} \quad \text{—(3)}$$

To eliminate intermediate $H_2NO_2^+$, apply S.S.A.

$$\frac{d[H_2NO_2^+]}{dt} = k_1[HNO_2][H^+] - k_{-1}[H_2NO_2^+] - k_2[Br^-][H_2NO_2^+] = 0$$

$$k_{-1}[H_2NO_2^+] + k_2[Br^-][H_2NO_2^+] = k_1[HNO_2][H^+]$$

$$[H_2NO_2^+](k_{-1} + k_2[Br^-]) = k_1[HNO_2][H^+]$$

$$[H_2NO_2^+] = \frac{k_1[HNO_2][H^+]}{k_{-1} + k_2[Br^-]} \quad \text{—— (4)}$$

The reaction rate is in the rate-determining step

$$\text{rate} = k_2[Br^-][H_2NO_2^+] \quad \text{—— (5)}$$

$$= k_2[Br^-]\frac{k_1[HNO_2][H^+]}{k_{-1} + k_2[Br^-]}$$

$$= \frac{k_1 k_2 [Br^-][H^+][HNO_2]}{k_{-1} + k_2[Br^-]}$$

In order to set the observed rate law. assume
$k_{-1} \ggg k_2[Br^-]$

$$\text{rate} = \frac{k_1 k_2}{k_{-1}}[Br^-][H^+][HNO_2]$$

$$\text{rate} = k[Br^-][H^+][HNO_2]$$

To apply S.T.A;

(i) write a differential rate law to the rate determining step

(ii) eliminate the concentration a any reaction intermediate in step(i) expression.

Pre - equilibrium

$$A \underset{k_2}{\overset{k_1}{\rightleftharpoons}} M + C$$

$$M + B \xrightarrow{k_3} \text{Products}$$

* The rate ~~of~~ intermediate is formed by a reversible process prior to the final step of the reaction. Therefore it is called pre-equilibrium.

$$-\frac{d[B]}{dt} = k_3[M][B] \quad \text{—(1)}$$

Applying S.T.A to M

$$\frac{d[M]}{dt} = k_1[A] - k_2[M][C] - k_3[M][B] = 0$$

$$k_1[A] = k_2[M][C] + k_3[M][B]$$

$$= [M](k_2[C] + k_3[B])$$

$$[M] = \frac{k_1[A]}{k_2[C] + k_3[B]} \quad \text{—(3)}$$

Substitution equation (2) in (1),

~~if $k_2[C] >>> k_3[B]$~~

$$\frac{-d[B]}{dt} = \frac{k_3 k_1 [A][B]}{k_2[C] + k_3[B]}$$

if $k_3[B] >>> k_2[C]$

$$\frac{-d[A]}{dt} = k_1[A]$$

if $k_2[C] >> k_3[B]$